

Dereje D. Jima

Senior Bioinformatics Scientist | Computational Genomics, Statistics & Multi-Omics

Human Health and Disease | Epidemiologic Research | Multi-omics

📍 Cary, NC, USA | 📞 (xxx)-xxx-xxxx | @ ddjima2014@gmail.com | 🌐 www.ddjima.com

🌐 LinkedIn: dereje-jima-msc-phd-5395ab8 | 🐙 GitHub: ddjima

🆔 ORCID: 0000-0002-7784-1612 | 📄 ResearchGate: Dereje-Jima

PROFESSIONAL SUMMARY

- ▶ Senior bioinformatics scientist with 20+ years in computational biology and bioinformatics, with deep expertise in statistics, epidemiologic data analysis, genomics, epigenomics, transcriptomics, and integrative multi-omics research.
- ▶ Scientific and technical leader who partners with 50+ investigators across three universities on multicenter and cohort studies, sequencing study design, reproducible analysis, interpretation of high-dimensional data, and delivery of publication-ready results.
- ▶ Expert in multivariable statistical analysis, R/Bioconductor, Python, Linux/HPC, and workflow development, with applications across cancer, immunology, infectious disease, toxicology, and environmental health.

EDUCATION

🎓 PhD, Bioinformatics

North Carolina State University | August 2022 -- May 2026

🎓 MS in Bioinformatics

Wageningen University | 2003 -- 2005

🎓 BS in Plant Science

Haramaya University | 1996 -- 2000

CAREER HIGHLIGHTS

- 📄 **76 peer-reviewed publications** across genomics, epigenomics, human disease, epidemiologic research, and biomedical data science.
- 👤 **Co-investigator on NIH-funded research**, including R01, R21, and CHHE Center projects.
- 📊 **Multicenter epidemiologic and cohort analysis**, including multivariable generalized linear model (GLM) and robust linear model (RLM) analyses for the Newborn Epigenetics Study (NEST) cohort.
- 🔧 **Creator of the Human Imprintome Catalog**, a public database-backed scientific resource deployed and maintained on Linux infrastructure.

PROFESSIONAL EXPERIENCE

Senior Research Scholar

North Carolina State University | Raleigh, NC

November 2023 -- Present

- ▶ Promoted from Research Scholar in recognition of sustained leadership in bioinformatics, multi-omics analysis, and cross-institutional research.
- ▶ Lead bioinformatics strategy as CHHE Liaison for 50+ investigators across three universities, guiding multicenter and cohort study design, sequencing analysis, statistical methodology, and interpretation of high-dimensional datasets.
- ▶ Direct bioinformatics components of multiple NIH-funded R01, R21, and CHHE Center projects as co-investigator, contributing to proposals, study design, analytical planning, and research execution.
- ▶ Establish analytical plans and review scientific reports and technical results to ensure rigor, reproducibility, and publication readiness.
- ▶ Created and maintain the Human Imprintome Catalog (humanicr.org), a public database-backed resource deployed on a DigitalOcean Linux server.

Research Scholar

North Carolina State University | Raleigh, NC

September 2015 -- October 2023

- ▶ Developed scalable, reproducible pipelines for genome-wide DNA methylation and RNA-seq analysis and integration of large multi-omics datasets.

- ▶ Led end-to-end analyses across toxicology, cancer genomics, immunology, epidemiology, and environmental health, from quality control and statistical modeling through biological interpretation.
- ▶ Analyzed multicenter epidemiologic and cohort data, including the Newborn Epigenetics Study (NEST) cohort, using multivariable generalized linear models (GLMs) and robust linear models (RLMs) to evaluate relationships among exposures, molecular measures, and health outcomes.
- ▶ Analyzed Genotype-Tissue Expression (GTEx) data for genomics and transcriptomics research, incorporating large-scale public reference data into reproducible analytical workflows.
- ▶ Enabled multi-institutional and large-cohort research through reproducible workflows, rigorous statistical analysis, and integration of epidemiologic, clinical, and molecular datasets.
- ▶ Strengthened funded translational research by contributing bioinformatics expertise to grant proposals, study protocols, and data-analysis plans.
- ▶ Translated analytical findings into publications, presentations, technical reports, and defensible scientific conclusions for interdisciplinary collaborators.

CNTS, Senior Bioinformatician

Walter Reed Army Institute of Research | Silver Spring, MD

June 2014 -- September 2015

- ▶ Conducted bioinformatics research in infectious disease, data integration, and application development for pathogen-focused sequencing studies.
- ▶ Redesigned a microbial pathogen discovery workflow and initiated a 16S metagenomics pipeline to improve analytical capability for surveillance-oriented research.
- ▶ Supported surveillance- and vaccine-related research through bioinformatics analysis, data integration, and scientific interpretation.
- ▶ Contributed genome assembly and downstream sequence analysis for *Candidatus Rickettsia asemboensis* strain NMRCii.
- ▶ Held U.S. Government Public Trust security clearance.

Bioinformatician I

Duke University Medical Center | Durham, NC

March 2007 -- June 2014

- ▶ Contributed to cancer genomics and B-cell biology research through computational analysis, scientific reporting, and manuscript development.
- ▶ Designed and implemented analytical frameworks for sequencing, gene expression, and epigenetic microarray studies.
- ▶ Analyzed biological data generated using diverse high-throughput platforms and translated results into publication-ready scientific outputs.
- ▶ Built, configured, and administered a Linux compute cluster with 16 Supermicro servers to support high-throughput bioinformatics workloads.

Research Associate

North Carolina State University | Raleigh, NC

September 2005 -- March 2007

- ▶ Conducted innate immunity research focused on natural killer cells.
- ▶ Performed independent research, expression microarray analysis, and molecular biology wet-lab work.
- ▶ Managed laboratory operations, equipment, and regulated biological materials.

Visiting Research Scholar

Oklahoma State University | Stillwater, OK

March 2005 -- September 2005

- ▶ Worked on database integration, parsing, Perl scripting, and MySQL-based data mining.

Assistant Lecturer

Ambo University | Ambo, Ethiopia

December 2000 -- August 2003

- ▶ Taught university-level courses and conducted independent academic research.

TECHNICAL PROFILE

 **Programming and Scripting:** R/Bioconductor, Python, shell/Bash, awk, sed, SQL, NoSQL, JSON; prior experience with Perl

 **Statistics and Epidemiology:** Multivariable analysis, generalized linear models (GLMs), robust linear models (RLMs), epidemiologic and cohort data analysis, statistical genomics, high-dimensional data analysis, observational study design, quality control, and biological interpretation

 **Platforms, Data, and Web:** Linux/HPC, Nextflow, Git, PostgreSQL, MySQL, SQLite, full-stack scientific web development, HTML/CSS, Bootstrap, WordPress, DigitalOcean deployment, Linux server administration

 **Data Modalities:** RNA-seq, Methyl-seq, ChIP-seq, FAIRE-seq, methylation arrays, exome sequencing, whole-genome sequencing, WGS, WES, Tempo-seq

 **AI and Computational Methods:** LLM-assisted scientific programming, retrieval-augmented generation (RAG), workflow automation, Google Colab, and reproducible AI-assisted analysis

HONORS AND AWARDS

 **PRIDE OF THE WOLFPACK**, recognition for contributions to North Carolina State University, March 2026.

 **CHHE Outstanding Member Award**, North Carolina State University, 2019.

 **Netherlands Fellowship Program**, scholarship support for MS study.

 **Degree with Distinction**, Haramaya University.

PROFESSIONAL DEVELOPMENT

 **Generative AI and RAG**, UNC-Chapel Hill/NCDS, 2026: three instructor-led courses covering generative AI for coders, AI foundations, and retrieval-augmented generation.

 **Nextflow Advanced Training and Nextflow Summit**, Seqera, Boston, MA, 2026.

 **Large Language Model Symposium**, North Carolina State University, 2025: hands-on LLM and machine-learning exercises in Google Colab.

PROFESSIONAL AFFILIATIONS AND SERVICE

 Member, Center for Human Health and the Environment (CHHE), North Carolina State University.

 Member, Bioinformatics Research Center (BRC), North Carolina State University.

 Affiliate Member, University of North Carolina at Chapel Hill, Chapel Hill, NC, 2016--Present.

 Member, International Society of Developmental and Comparative Immunology, 2005--2007.

 Former editorial board member, Austin Medical Sciences Journal.

 Peer reviewer for genomics and biomedical journals, 3+ manuscripts annually.

SELECTED PUBLICATIONS

- D. D. Jima, D. A. Skaar, A. Planchart, A. Motsinger-Reif, S. E. Cevik, and S. S. Park, "Genomic map of candidate human imprint control regions: The imprintome," *Epigenetics*, vol. 17, no. 13, pp. 1920–1943, 2022.
- J. Zhang et al., "The genomic landscape of mantle cell lymphoma is related to the epigenetically determined chromatin state of normal b cells," *Blood*, vol. 123, no. 19, pp. 2988–2996, 2014.
- J. Zhang, V. Grubor, C. L. Love, A. Banerjee, K. L. Richards, and P. A. Mieczkowski, "Genetic heterogeneity of diffuse large b-cell lymphoma," *Proceedings of the National Academy of Sciences*, vol. 110, no. 4, pp. 1398–1403, 2013.
- C. Love et al., "The genetic landscape of mutations in burkitt lymphoma," *Nature Genetics*, vol. 44, no. 12, pp. 1321–1325, 2012.
- D. D. Jima, J. Zhang, C. Jacobs, K. L. Richards, C. H. Dunphy, and W. W. L. Choi, "Deep sequencing of the small rna transcriptome of normal and malignant human b cells identifies hundreds of novel micrnas," *Blood*, vol. 116, no. 23, e118–e127, 2010.
- J. Zhang et al., "Patterns of microrna expression characterize stages of human b-cell differentiation," *Blood*, vol. 113, no. 19, pp. 4586–4594, 2009.

LINKS TO FULL PUBLICATIONS

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